

FakeFormer: Transformer-Based Lightweight Deepfake Video Detection Model

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Abstract—In recent years, deepfake videos become increasingly realistic, making them nearly indistinguishable from the naked eye. The misuse of these deepfake videos poses a major threat to information security, leading researchers to develop effective deepfake video detection models. Video temporal features often contain rich and valuable information for identifying the authenticity of videos. Given that transformers have proven highly effective at modeling these features, researchers have been prompted to develop transformer-based detection models. However, due to the high-dimensional nature of video data, the efficiency of such models is often compromised, making it a challenging problem. In this paper, we propose a specialized transformer-based model for deepfake video detection, called FakeFormer. FakeFormer is a lightweight video-level detection model that maintains high efficiency, despite using a sequence of video frames as direct input. Additionally, FakeFormer fully integrates the strengths of convolutional neural networks and transformers, enabling it to effectively model both the temporal and spatial features of videos to accurately assess their authenticity. Numerous experiments validate the effectiveness of FakeFormer, including intra-dataset evaluation, cross-dataset evaluation and ablation study.

Index Terms—Deepfake detection, Spatio-temporal feature, Lightweight video transformer

I. INTRODUCTION

Deepfake is the technique of swapping source and target faces to create manipulated images and videos. With the advancement of deep learning techniques, deepfake techniques have seen significant breakthroughs. However, the popularity of deepfakes has also created a hidden danger, leading to an increasing number of real-world cases suffering from the harm caused by forged videos. As a result, developing effective deepfake detection technology has become both a valuable and urgent need.



Fig. 1. Frames of deepfake videos.

Deepfake video detection is essentially a binary classification task that distinguishes between real and fake videos. Existing methods can be broadly categorized into image-level and video-level approaches. Image-level methods [1], [2] treat each

video frame independently, analyzing them individually and aggregating the results for a final judgment. However, these methods only utilize spatial information, ignoring temporal inconsistencies in forged videos, as illustrated in Fig. 1. Video-level detection methods [3], [4] add attention to these inconsistencies, enhancing detection performance. However, the inherent high dimensionality of video data makes extracting temporal features a challenging task. Additionally, although transformers have become the first choice for sequential tasks, they are inefficient when applied to video data. This problem is further accentuated by the addition of the high computational complexity associated with the self-attention operation. At the same time, using recurrent neural networks [5] suffers from a lack of detection performance, making it difficult to find an ideal balance.

To address the aforementioned problems, we propose a lightweight and effective transformer-based deepfake video detection model, named FakeFormer. Inspired by a study exploring the core components of transformers, the authors [6] discover that the strong performance of transformers is mainly due to their four-stage architecture, achieved by replacing self-attention with parameter-free pooling operations. FakeFormer is also based on the four-stage architecture and further combines the strengths of convolutional neural networks [7] and transformers. It takes continuous video frames as direct input, effectively and cohesively extracting spatial features from the video frames and temporal features from the entire video, leveraging both for deepfake detection. Extensive experiments on several public datasets demonstrate that our method effectively detects deepfake videos, resulting in rich performance advantage over existing methods.

II. THE PROPOSED FAKEFORMER MODEL

A. Motivation

Transformer's powerful ability to extract temporal features aligns well with the needs of deepfake video detection tasks, driving researchers to develop transformer-based detection models. However, video data introduce a temporal dimension to image data and need to be adapted for the application of the visual transformer. The most intuitive approach is to design a 3D transformer, but we find that this model results in extremely high computational complexity, making it difficult to train.

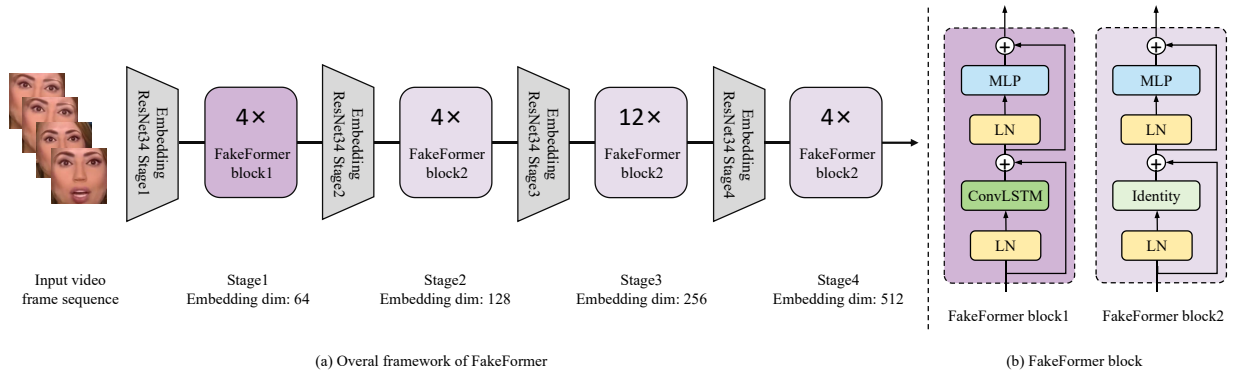


Fig. 2. (a) The overall framework of FakeFormer. In FakeFormer, stages [1, 2, 3, 4] consist of [4, 4, 12, 4] blocks, respectively. The embedding dimensions of each stage are [64, 128, 256, 512]. (b) The structure of the two basic blocks in FakeFormer, which are applied to the first stage as well as to the other stages, respectively.

We test it with two 3090 GPUs, and it takes several hours to complete one epoch. Another class of methods employs a convolutional model to extract spatial features, followed by a transformer to capture temporal features. These two-stage approaches are similarly inefficient. To better apply transformer to deepfake video detection, we design a lightweight video-level detection model FakeFormer, based on the core component of transformer. The overall architecture is shown in Fig. 2. FakeFormer follows the four-stage design guideline and directly uses continuous video frame sequences as input. It can simultaneously extract the spatio-temporal features from the video and provide the discrimination results without additional processing.

B. Embedding Layer

Although deepfake techniques continue to mature, forged video frames still contain subtle imperfections that can be detected by well-trained models. These spatial inconsistencies are crucial for distinguishing real from fake videos, requiring the model to effectively capture spatial details. In our preliminary experiments, we find that relying solely on the transformer’s four-stage architecture, results in poor performance in the deepfake video detection task. And this issue persists even when fine-tuning with a pre-trained version on a large-scale dataset. We analyze that it is caused by the model’s inability to capture spatial details. Subsequently, we find that adding two simple convolutional layers before the first stage significantly improves the model’s performance, confirming our hypothesis.

In the preliminary experiments, we use a single video frame as input, where the input to the embedding layer consists of image patches. In our approach, we use a sequence of consecutive video frames as input, where the embedding layer receives images. Since the original embedding module is a linear mapping and is no longer suitable, it has been removed. To compensate for the loss of the original embedding module and more effectively capture spatial details in the video frames, we introduce multi-layer convolution before each stage to perform the embedding operation. Rather than manually designing the convolutional layers, we employ a

lightweight convolutional model, ResNet34, and divide it into four stages [64, 128, 256, 512] according to the dimensions of the resulting feature space, which are inserted before each corresponding stage of our model.

C. Basic block

In real videos, the spatial details in the frames remain consistent throughout the sequence, whereas in fake videos, they show inconsistency. This temporal variation in spatial details is crucial for deepfake detection. After applying the embedding operation to the video frames, the resulting feature space is relatively large. Using self-attention to extract temporal features requires flattening the feature space, which will lead to the loss of valuable spatial information and further impact the extraction of effective temporal inconsistency cues. To address this, we design a ConvLSTM module to replace self-attention, enabling more efficient temporal feature extraction from video frame sequences. ConvLSTM is a two-layer network, with both the input and hidden dimensions matching the embedding dimension. It uses a 3x3 convolutional kernel and applies the hidden state of the second layer at each time step as the output. In our design, the ConvLSTM module simultaneously computes the input gate i_t , forget gate f_t , output gate o_t , and candidate state $\tilde{C}_t$ through a single convolutional operation. Specifically, the input feature X_t and the hidden state H_{t-1} from the previous time step are concatenated to form a new tensor Z_t :

$$Z_t = \text{Concat}(X_t, H_{t-1}). \quad (1)$$

Assuming the number of channels for X_t and H_{t-1} is C , the number of channels for Z_t is $2C$. We then apply a convolutional operation with an input channel number of $2C$ and an output channel number of $4C$ to generate a tensor G_t with $4C$ channels:

$$G_t = \text{Conv}(Z_t). \quad (2)$$

Then, G_t is split along the channel dimension into four groups, each containing C channels, which are used to com-

pute the input gate i_t , forget gate f_t , output gate o_t , and candidate state $\tilde{C}_t$, respectively. The formulas are as follows:

$$\begin{aligned} i_t &= \sigma(G_t^{0:C-1}), \\ f_t &= \sigma(G_t^{C:2C-1}), \\ o_t &= \sigma(G_t^{2C:3C-1}), \\ \tilde{C}_t &= \tanh(G_t^{3C:4C-1}), \end{aligned} \quad (3)$$

where $0 : C - 1$ represents the first C channels, and the others follow accordingly.

Finally, the cell state and hidden state are updated based on the gating signal and candidate state:

$$\begin{aligned} C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \\ H_t &= o_t \odot \tanh(C_t), \end{aligned} \quad (4)$$

where $\odot$ denotes the element-wise multiplication.

The temporal inconsistency required to identify forged videos is often hidden in the shallow features of the video frame sequence. However, the higher stages of the model extract global high-level semantics, such features exhibit relatively high consistency throughout the sequence, regardless of whether the video is real or fake. Therefore, we use ConvLSTM only in the first stage of the model to extract temporal inconsistency cues. We further prove our point by comparing the positions where ConvLSTM is applied in the ablation experiments. Finally, for the other stages of the model, we use identity, a parameter-free operation to reduce the overall complexity of the model.

III. EXPERIMENT

In this section, we demonstrate the effectiveness of our method through experimental results, including intra-dataset evaluation, cross-dataset evaluation, ablation study and visualization experiment.

A. Datasets

We validate our model using three publicly available datasets: FaceForensics++ [2], Celeb-DF [8] and DFDC [9]. FaceForensics++ (FF++) contains 1000 real videos from YouTube and 4000 fake videos generated using four forgery methods: DeepFakes (DF), FaceSwap (FS), Face2Face (F2F), and NeuralTextures (NT). FF++ offers three versions with different compression rates (raw, c23, c40). In our experiments, we use the low-compression version (c23, HQ). Celeb-DF is a high-quality dataset containing 890 real videos and over 5000 fake videos, which we process using data equalization techniques. DFDC consists of 400GB videos and is a challenging dataset, filled with various types of noise, making it more difficult to identify the authenticity of the videos.

B. Implementation Details

In the data preprocessing phase, we divide each video into eight segments and select four consecutive frames from each. We then use the open-source dlib [10] face detector to crop the face regions and resize each image to 224x224. For each dataset, we split it into training, validation, and testing sets with an 8:1:1 ratio.

We use the Adam optimizer with a learning rate of $1e^{-4}$ and a batch size of 40. The learning rate is halved every 5 epochs during training, and the total number of epochs is 50. The loss function is the binary cross-entropy loss. Evaluation metrics include accuracy (ACC) and the area under the ROC curve (AUC).

C. Intra-Dataset Evaluation

Intra-dataset evaluation refers to training and testing on homologous data to assess the model's fundamental expressive power, which is its ability to extract forgery clues from deepfake videos. We perform comparisons on the four forgery methods in FF++ as well as Celeb-DF, with the results shown in Table. I. Xception [11] is a generalized vision model and is not specifically designed for deepfake detection, and it performs the worst among the evaluated methods. In contrast, the other models are specifically designed for deepfake video detection tasks and show varying degrees of performance advantages over Xception. Furthermore, our model achieves the best performance on each dataset, which substantiates the effectiveness of the proposed approach.

TABLE I
INTRA-DATASET EVALUATION RESULTS. WE REPORT THE ACC (%) OF SEVERAL DEEFAKE DETECTION METHODS ON THE FACEFORENSICS++ AND CELEB-DF DATASETS. BOLD REPRESENTS THE BEST RESULTS.

Method	DF	F2F	FS	NT	Celeb-DF
Xception [11]	91.37	90.28	91.54	88.49	85.19
CNN-RNN [5]	94.63	92.28	93.15	89.02	91.75
3DCNN [4]	95.88	91.21	92.67	90.30	94.37
ADD [3]	94.12	93.09	94.25	88.77	92.74
RE-C [1]	96.06	95.48	97.33	93.52	95.90
UCT [12]	97.83	97.05	96.68	92.74	96.13
FakeFormer	98.35	97.12	97.86	94.40	97.29

D. Cross-Dataset Evaluation

To verify the generalization ability of FakeFormer, we train it on FF++ and test it on Celeb-DF and DFDC. We compare the performance of several models, and the results are presented in Table. II. Although the models have never seen the data from the two test datasets during training, there are still some common issues across different deepfake videos, such as defects in details. Therefore, after training, all the models maintain varying degrees of generalization ability. Additionally, it can be seen that our model achieves the best results, indicating that our model can more effectively mine the common spatio-temporal inconsistency defects existing in forged videos, thereby distinguishing between real and fake videos more efficiently.

E. Ablation Study

We conduct experiments on Celeb-DF to evaluate the impact of different temporal feature extraction modules and their placement on detection performance. The results are shown in Table. III. The model without any temporal module, relying solely on the identity operation, performs the worst. Introducing a temporal feature extraction module in the first stage

TABLE II
CROSS-DATASET EVALUATION RESULTS. ALL THE MODELS ARE TRAINED ON FF++ THEN TESTED ON DFDC AND CELEB-DF. THE EVALUATION METRIC IS AUC (%), WITH BOLD INDICATING THE BEST RESULTS.

Methods	Training dataset	Test datasets		
		Celeb-DF	DFDC	Cross-Avg
Xception [11]	FF++	64.2	59.4	62.30
CNN-RNN [5]	FF++	65.8	60.3	63.05
3DCNN [4]	FF++	67.4	62.8	65.10
ADD [3]	FF++	67.9	61.4	64.65
RE-C [1]	FF++	69.3	66.7	68.00
UCT [12]	FF++	70.5	64.1	67.30
FakeFormer	FF++	72.1	67.8	69.95

significantly enhances the model’s performance. Moreover, the ConvLSTM module achieves better performance than the self-attention module, indicating that effectively preserving spatial features facilitates temporal information extraction. Additionally, adding ConvLSTM modules to the other three stages does not yield better results than using it in the first stage. Finally, incorporating ConvLSTM in both the first and second stages leads to a 0.05% performance improvement compared to using it only in the first stage. However, considering the trade-off between computational cost and performance, we ultimately use the ConvLSTM module only in the first stage.

TABLE III
ABLATION STUDY RESULTS. WE REPORT THE ACC (%) ON THE CELEB-DF DATASETS.

Temporal module	Adding position	Celeb-DF
Identity	First stage	94.92
Self-attention	First stage	96.57
ConvLSTM	First stage	97.29
ConvLSTM	Second stage	97.04
ConvLSTM	Third stage	95.78
ConvLSTM	Fourth stage	95.13
ConvLSTM	First and second stages	97.34

F. Visualization

To intuitively illustrate the rationale behind our method’s judgment in detecting fake videos, we use Grad-CAM [13] to generate activation maps and present the results using the FF++ dataset as an example. As shown in Fig. 3, the first row represents the video frames input to the model, while the second row shows their corresponding activation maps. Additionally, the column labeled YouTube represents real video frames, while the remaining four columns correspond to the four forgery methods in FF++. It can be seen that the model accurately identifies forgeries, as indicated by the red regions in the activation maps, across all four forgery methods. In contrast, no forged regions are detected in real videos.

IV. CONCLUSION

In this paper, we propose FakeFormer, a lightweight and effective video transformer model for deepfake detection. FakeFormer leverages the core strengths of convolutional neural networks and transformers. It takes a continuous sequence of video frames as the direct input, fully extracting valuable

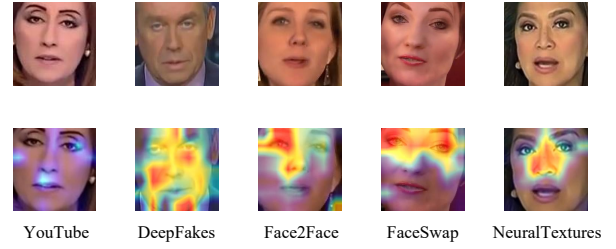


Fig. 3. Visualization results of the Grad-CAM outputs.

temporal and spatial features to produce authenticity identification results for the video. Experiments demonstrate that FakeFormer outperforms most existing methods for deepfake video detection. In future work, we will focus on improving its generalization to address potential future forgeries.

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